

To design a low-power two-stage Miller-compensated Operational Transconductance Amplifier (OTA) in GF180MCU technology that meets the specified DC gain, unity-gain bandwidth, phase margin, slew rate, load capacitance, and power consumption requirements through appropriate transistor sizing, gm/ID-based biasing, frequency compensation, and load optimization.

Technology

GF180MCU PDK

TT Corner

Temperature = 27°C

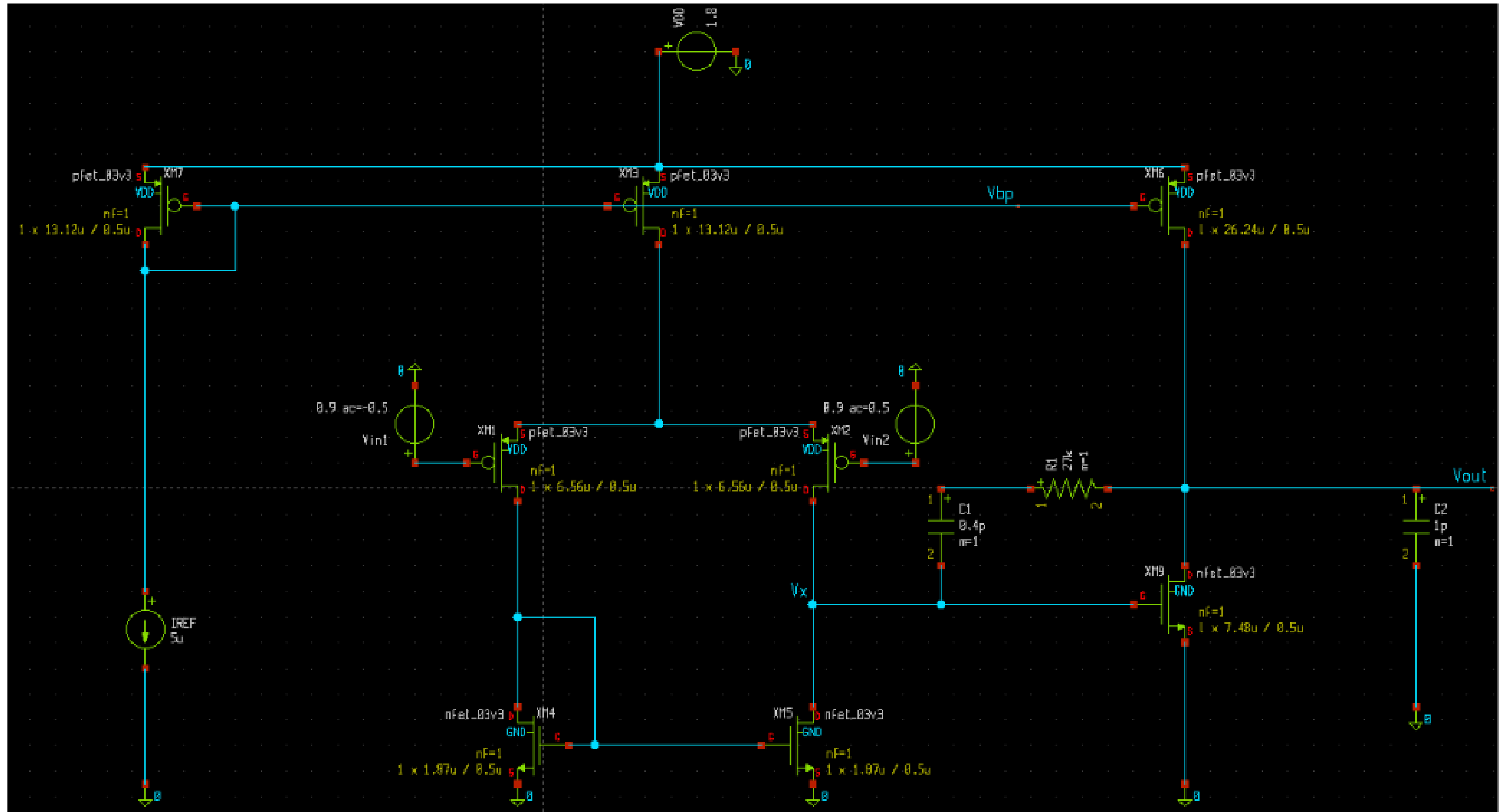
Supply

VDD = 1.8 V

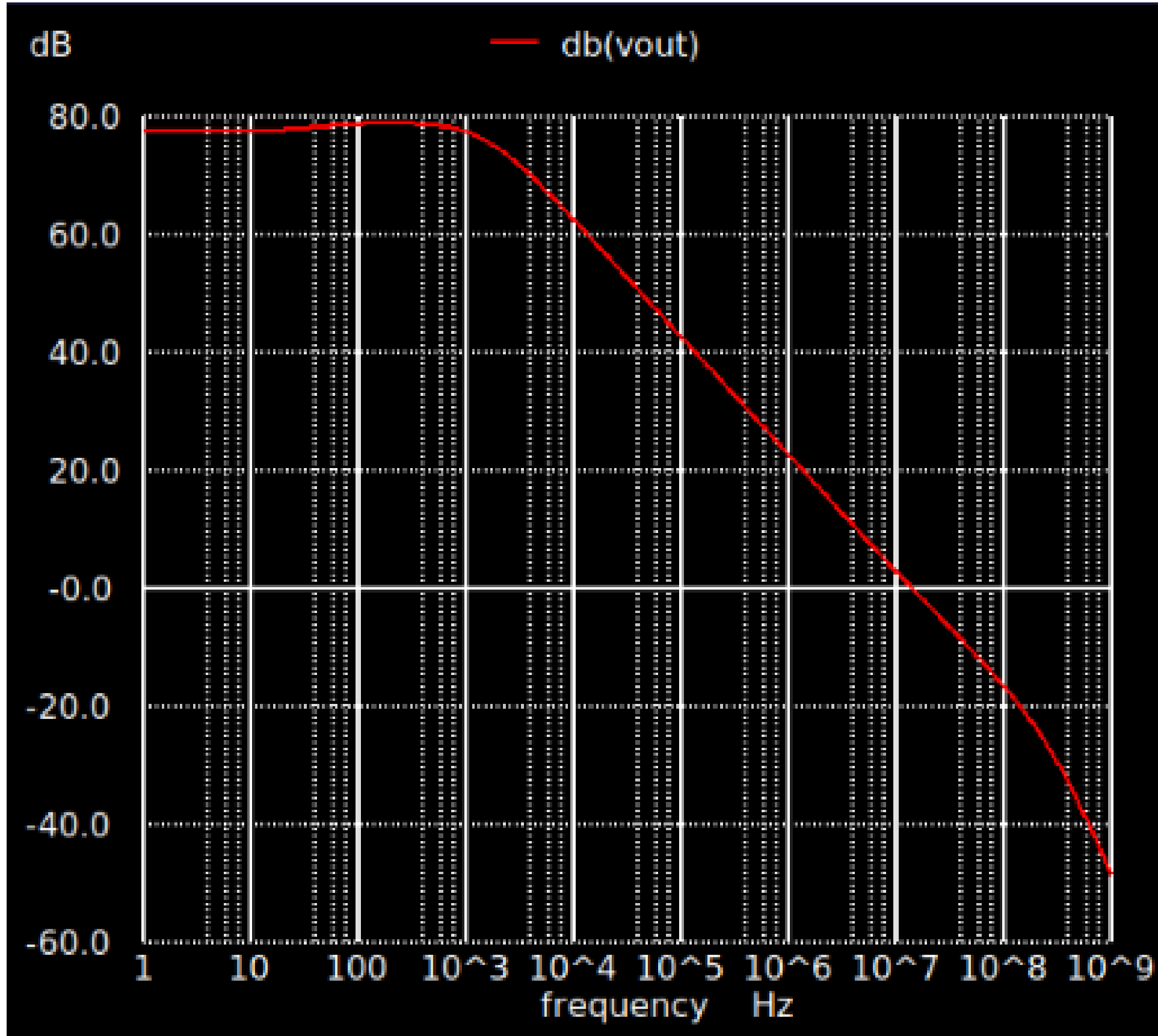
Specifications

- DC Gain (A_{vo}) ≥ 60 dB
- Unity Gain Bandwidth (UGB) ≥ 10 MHz
- Phase Margin (PM) $\geq 60^\circ$
- Load Capacitance (CL) 1 pF
- Power Consumption ≤ 100 μ W
- Slew Rate (SR) ≥ 5 V/ μ s

Schematic

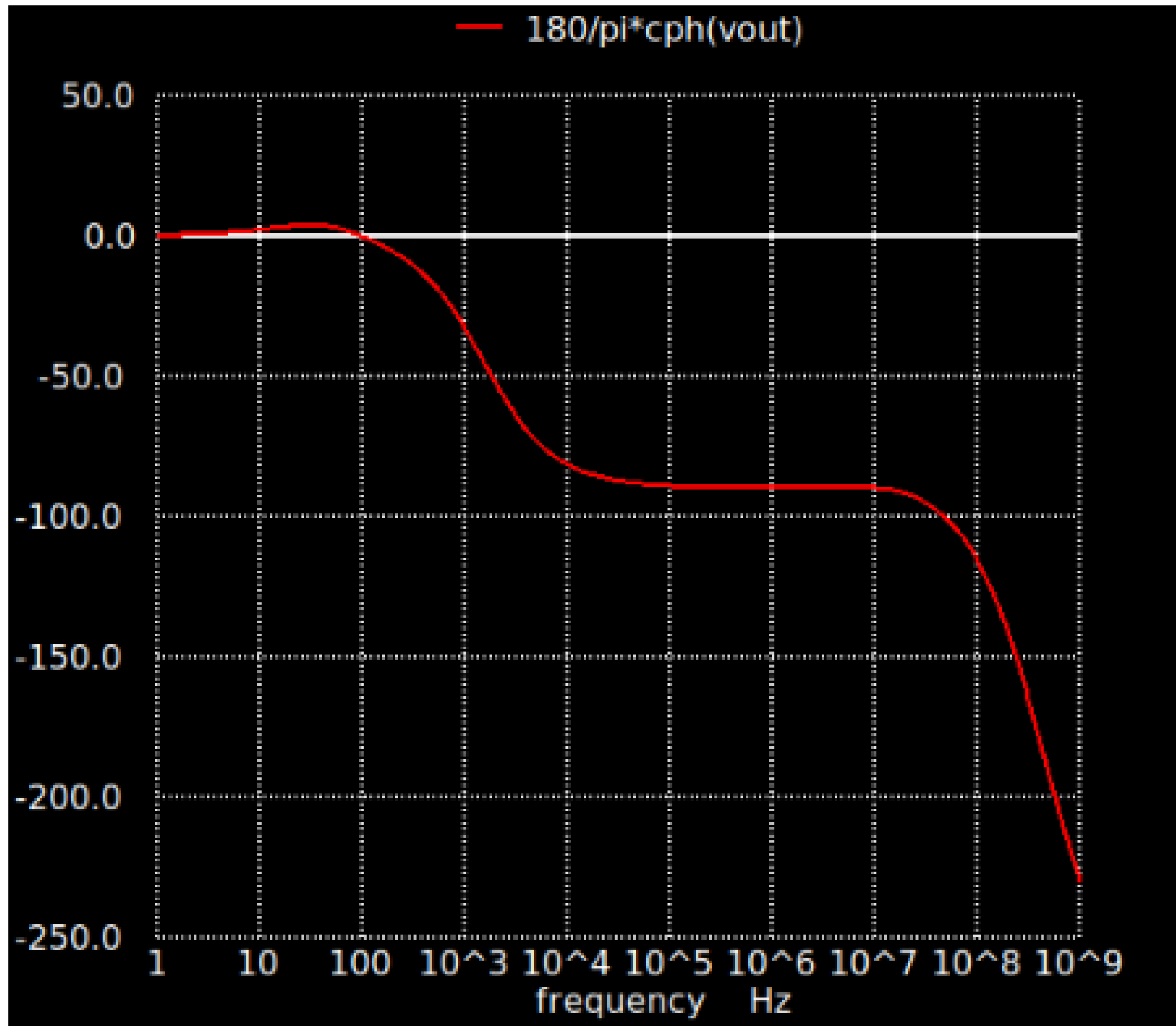


Gain & BandWidth Verifiatiion



Gain :78 db
UGB :10.2 Mhz

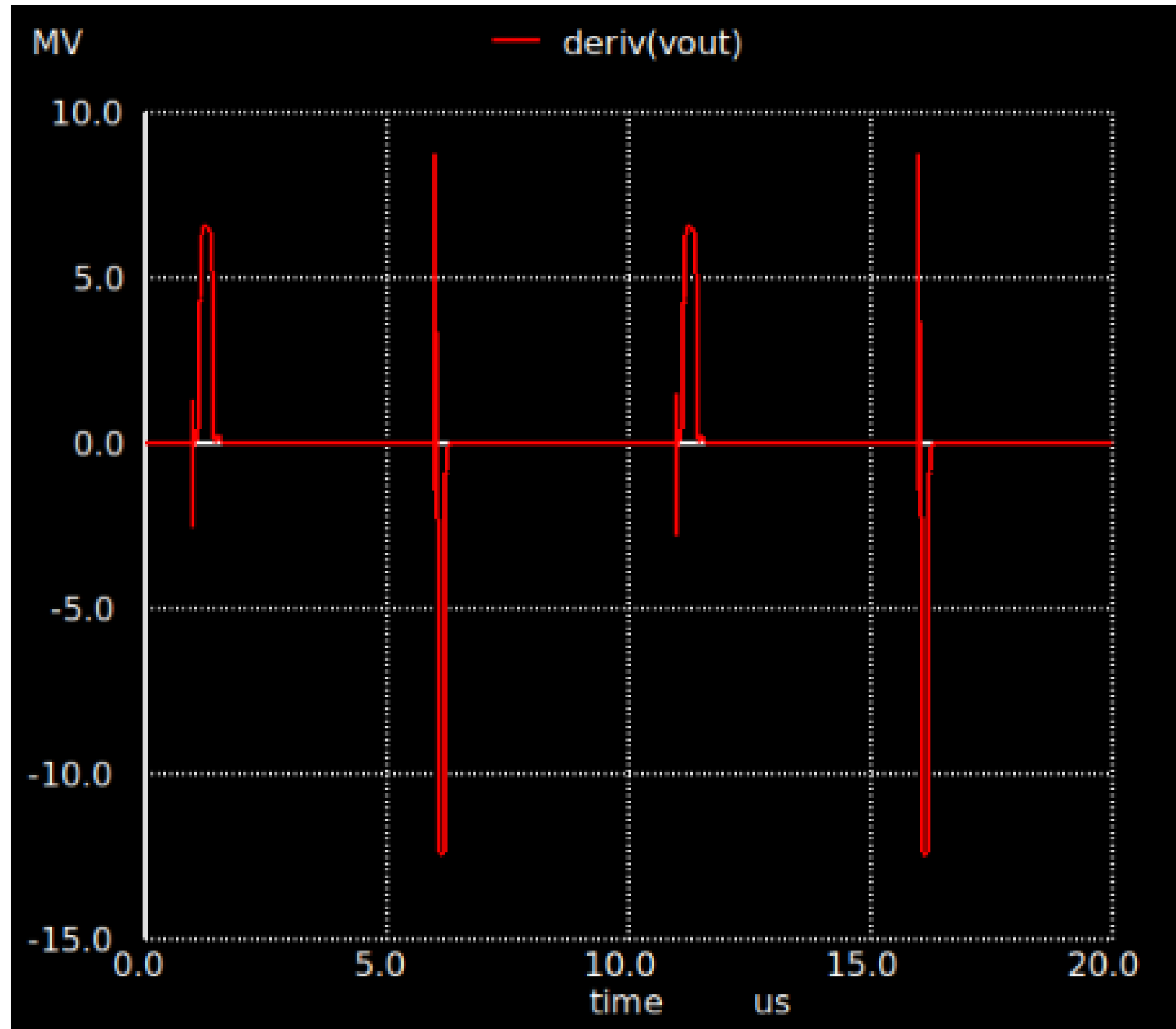
Phase Margin Calc



Phase @ UGB $\approx -88^\circ$

$\Rightarrow$ PM $\approx 92^\circ$

Slew rate verification



$$\text{SR}^+ \sim 6.5\text{V}/\mu\text{s}$$

$$\text{SR}^- \sim 12.5\text{V}/\mu\text{s}$$

Specification Verification

- ✓ DC Gain = 77.6 dB (Requirement: ≥ 60 dB)
- ✓ Unity-Gain Bandwidth (UGB) = 10.53 MHz (Requirement: ≥ 10 MHz)
- ✓ Phase Margin (PM) = 64.7° (Requirement: $\geq 60^\circ$)
- ✓ Power Consumption = 34.7 μ W (Requirement: ≤ 100 μ W)
- ✓ Positive Slew Rate (SR+) ≈ 6.5 V/ μ s (Requirement: ≥ 5 V/ μ s)
- ✓ Negative Slew Rate (SR-) ≈ 12.5 V/ μ s
- ✓ Load Capacitance = 1 pF (Requirement: 1 pF)
- ✓ Supply Voltage = 1.8 V
- ✓ Miller Compensation with 27 k Ω Nulling Resistor Successfully Achieved Stability Enhancement
- ✓ All MOSFETs Operate in Saturation Region at the Nominal Operating Point